1 Small Language Models and the Future of Agentic AI [1]

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Small Language Models are sufficiently powerful, inherently more suitable and necessarily more economical for many invocations in agentic systems and therefore the future of Agentic AI.

Agentic AI Sector had seen more than 2 billion US Dollars in Startup Funding and this number is expected to grow to 200 Billion by 2034

1.1 Small Language Models

A Small Language Model is a Language Model that can fit onto a common consumer electronic device and perform inference with latency sufficiently low to be practical when serving the agentic requests of one user.

As of 2025, we would be comfortable with considering most models below 10 billion parameters in size of be SLMs,

Agent: Emphasizing the software with some agency as a whole

Agent System: highlighting the systems aspect of the agentic application as a sum of its components

SLMs are principally sufficiently powerful to handle language modeling errands of agentic applications. SLMs Inherently more operationally suitable for use in agentic systems than LLMs. They are necessarily more economical for vast majority of Language Models uses in agentic systems than their general purpose Large Language Models couterparts by the virtue of their smaller size.

1.1.1 SLMs are sufficiently powerful to take the place of LLMs in agentic systems.

- **Microsoft Phi Series**: Phi-2 (2.7 Billion) achieves commonsense reasoning scores and code generation scores on par with 30 Billion models while running 15 times faster
- **NVIDIA Nemotron H Family**: hybrid Mamba Transformer models achieves instruction following and code generation accuracy comparable to dense 30 billion LLMs
- **Huggingface SmolLM2 seires**: sizes ranging from 125 million to 1.7 billion parameters each run up in their language understanding, tool calling and instruction following performance to 14 billion contemporaries.
- **NVIDI Hymba 1.5B**: Mamba attention hybrid head SLM deomnstrates best instruction accuracy and 3.5 times greater token throughput than comparably sized transformer models.
- **DeepSeek R1 Distill series**: trained on samples generated by DeepSeek R1 demonstrating strong commonsense reasoning capabilities. DeepSeek R1, Distill Qwen 7B outperforms large proprietary models
- **DeepMind Retro 7.5B**: Retrieval Enhanced Transformers (RETRO) is a 7.5 Billion parameter model augmented with an expensive external text database achieving performance comparable to GPT-3.
- **Salesforce xLAM-2-8B**: 8 Billion model achieves state of the art performance in tool calling despite is relatively modest size surpassing frontier models.

Reasoning capabilities of Small Language Models can be enhanced by light weight selective fine tuning. Toolformers (6.7 Billion) outperforms GPT-3 (175 Billion)

1.1.2 SLMs are more economical in agentic systems

- Inference Efficieny
- Fine Tuning Agility
- Edge Deployment
- Parameter and embedding space utilisation

1.1.3 SLMs posess greater operational flexibility in comparision to LLMs

more affordable and practical to train, adapt, and deploy multiple specialised expert models for different agentic routines

1.1.4 Agentic Applications are interfaces to a limited subset of langauge model capabilities

An AI agent is essentially a heavily instructed and externally choreographed gateway to a language model featuring a human computer interface and a selection of tools that when engaged correctly do something of utility. SLMs when appropiately fine tuned for selected prompts would suffice while having the benefits of increased efficiency and greater flexibility.

1.1.5 Agentic Interactions necessitate close behavioral alignment

SLM trained with a single formatting decision enforced during its post training or encouraged through additional fine tuning at a low cost is preferable over a general purpose LLM in the context of AI agents

1.1.6 Agentic systems naturally allow for heterogeneity in the selection of models that they use

All language models could in principle be specialised SLMs- one for conversationality, another one for carrying out controller defined language modelling tasks

1.2 Barriers in Adoption of SLMs

1.2.1 Large amounts of upfront investments in centralised LLM inference infrastructure

Large capital bets have been made on centralised LLM inference being the leading paradigm in providing AI services in the future

1.2.2 Use of generalist benchmarks in SLM training, design and evaluation

If one focuses solely on benchmarks in their measuring the agentic utility of agents, the studied SLMs easily outperform larger models.

1.2.3 Lack of popular awareness

can be broken if the economic benefits of the SLMs are better known

1.3 LLM to SLM Conversion Algorithm

The following simplified algorithm outlines a practical process for transitioning from large language model (LLM) generalists to small language model (SLM) specialists:

1. **Collect Usage Data:** Securely log agent interactions, including inputs, outputs, and tool calls, ensuring encryption and anonymization of all data.
2. **Clean and Filter Data:** Gather sufficient examples (typically 10k–100k) and remove or mask sensitive information. Optionally, paraphrase domain-specific content to preserve privacy.
3. **Identify Task Clusters:** Apply unsupervised clustering to group similar interactions or operations. Each cluster represents a potential specialization area (e.g., summarization, data extraction, code generation).
4. **Select Suitable SLMs:** Choose smaller models that best fit each identified task based on performance, efficiency, and deployment requirements.
5. **Train Specialized SLMs:** Fine-tune the selected SLMs on their respective datasets using parameter-efficient methods such as LoRA or QLoRA. Optionally, apply knowledge distillation from the LLM to transfer nuanced capabilities.
6. **Iterate and Improve:** Periodically retrain and update both SLMs and the routing mechanism with new data to maintain accuracy and adapt to evolving usage patterns.

2 A Comprehensive Survey of Small Language Models in the Era of Large Language Models: Techniques, Enhancements, Applications, Collaboration with LLMs, and Trustworthiness [2]

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2.1 Introduction

Challenges in LLMs due to large parameter sizes and computational demands requiring cloud API raising privacy concerns limits real-time applications on edge devices, and increases fine-tuning costs.

Perform poorly in healthcare and law due to insufficient domain-specific knowledge, necessitating specialized models.

SLM favours low inference latency, cost-effectiveness, efficient development, and easy customization and adaptability particularly well-suited for resource-limited environments and domain knowledge acquisition thereby requiring localised data handling for privacy minimal inference latency for efficiency adnd domain knowledge acquisition through lightweight fine tuning.

Evolution of neural language models (BERT) - pretraining and fine tuning paradigm to T5's pre training + prompting approach and finally GPT 3's pre training + in context learning has greatly enhanced NLP

Definition 1. ***Neural Scaling*** *refers to the empirical relationship between a neural model's performance and the amount of compute, data, and model parameters used during training. It captures how increasing these resources leads to predictable improvements in performance, typically following a power-law trend.*

Formally, neural scaling laws express how a performance metric L (e.g., loss or error rate) decreases as a function of model parameters N, dataset size D, and compute budget C:

$$L(N, D, C) \propto N^{-\alpha_N} D^{-\alpha_D} C^{-\alpha_C}$$

where $\alpha_N, \alpha_D, \alpha_C$ are empirically determined scaling exponents.

These relationships enable researchers to predict the impact of scaling model size, data, or compute on performance and to identify diminishing returns where further scaling becomes inefficient.

With neural scaling, these models unpredictably exhibit sudden acquisition of versatile abilities termed as "Emergent Ability"

For instance, Llama-3.1 model with 405 billion parameters performs better in dialogue, logical reasoning, and programming compared to the smaller 7B counterpart

2.1.1 Disadvantages of current LLMs

- Large Computational power requirements leading to high costs
- LLMs are generally hosted in the cloud and used via cloud-based APIs due to the large GPU memory and computational cost
- Driven by personal agents, on-device deployment is a critical requirement
- Their large parameter count can cause inference delays from seconds to minutes
- To boost performance in specialized domains such as healthcare and law
- deploying general-purpose LLMs would be a waste of resources and such LLMs often cannot match the performance of models tailored for specific tasks

2.1.2 Advatanges of SLMs

- SLMs excel in efficiency, cost, flexibility, and customization.
- Provide significant computational savings in pre-training and inference with reduced memory and storage needs
- effective in resource-limited settings
- on-device processing by enhancing privacy, security, response times, and personalization

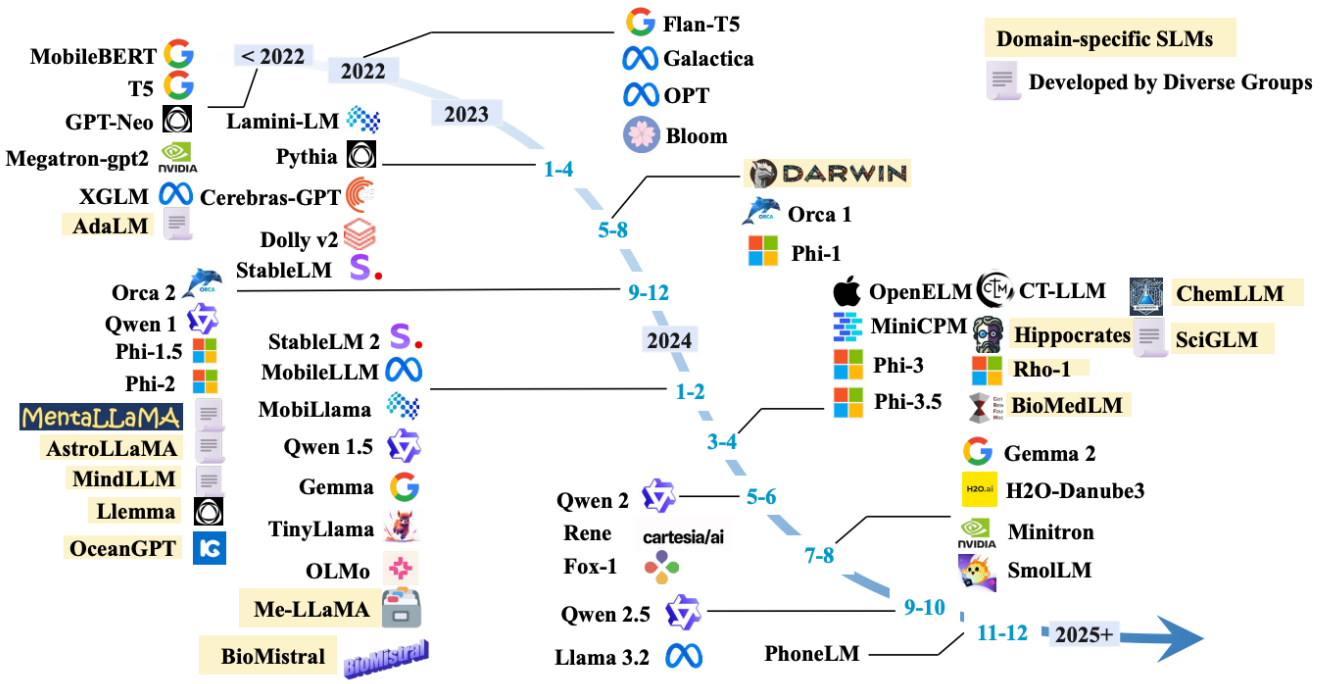

Fig. 3. A timeline of existing small language models.

Figure 1: Existing Small Language Models

2.2 Definition of LLMs

Some define SLMs as models with fewer than one billion parameters

Given specific tasks and resource constraints, we define SLMs as falling within a range where the lower bound is the minimum size at which the model exhibits emergent abilities for a specialized task, and the upper bound is the largest size manageable within limited resource conditions.

Resource effcient techniques: Quantisation aware training

2.3 Small Language Models

MobileBERT, T5, GPT-Neo, Megatron-gpt2, XGLM, AdaLM. Orca2, Qwen, Phi-1.5

2.4 Architecture of Small Language Models

Commonly employ Transformer Architecture which utilises self attention mechanism to manage long range text dependencies for maintaining performance with constrained resources at the cost high computational cost

Several subquadratic-time architectures such as Mambam, Hymba and xLSTM are proposed.

2.5 Transformer Architecture

Transformer Architecture: self attention mechanism allows language models to efficiently capture contextual information across longer sequences. generally adopts encoder-decoder structure featuring self attention mechanisms, feedforward networks, positional embeddings and layer normalisation.

2.5.1 Components of Transformer Architecture

Self Attention Mechanism: enables the model to evaluate the important of the tokens relative to each other. The self-attention mechanism is defined by the following equation:

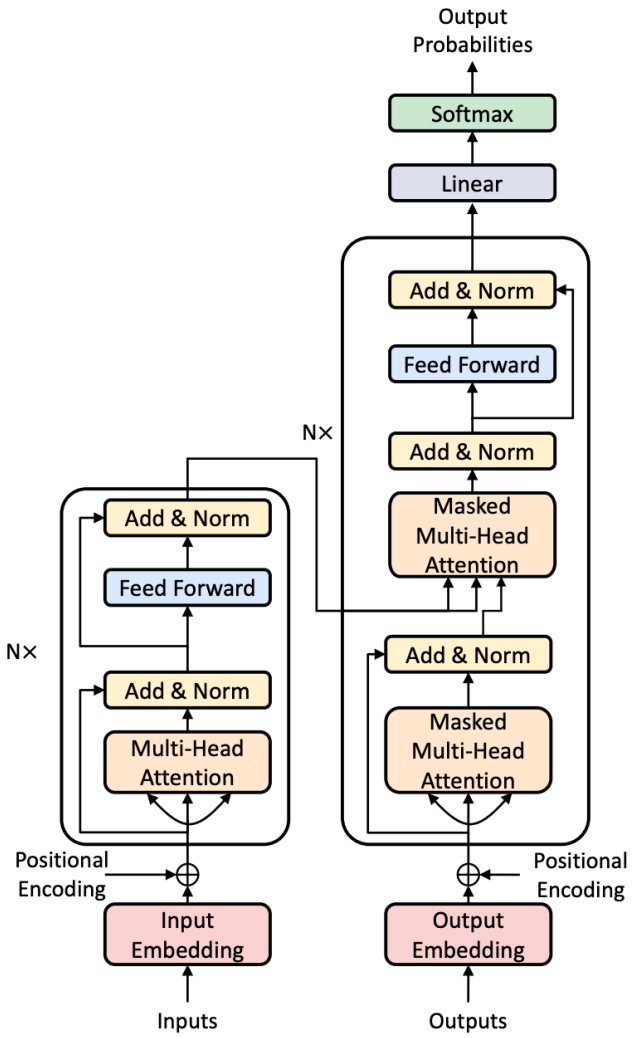

Figure 2: Transformer Architecture

$$Attention(Q, K, V) = softmax\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K, and V are query, key, and value matrices, scaled by $\sqrt{d_k}$ for stability where d_k is the dimension of key matrices. The dot product $QK^\top$ reflects the similarity between the query and key vectors.

Multi Head Attention: is the first method that uses multiple heads to capture diverse information. It allows the model to attend to different parts of the input sequence using multiple attention heads. Each head in the Multi-Head Attention mechanism operates independently, allowing the model to capture diverse aspects of the data: MHA, Multi-Query Attention (MQA) proposes that all attention heads share the same set of keys and values, which reduces the memory and computational overhead associated with storing and managing multiple key-value pairs. Grouped Query Attention (GQA) serves as a middle ground between MHA and MQA. It introduces subgroups of query heads (fewer than the total number of attention heads), where each subgroup shares a single key and value head. Several modifications have been introduced to further optimize self-attention mechanisms for specific challenges such as memory efficiency and computational speed. Unlike MQA and GQA, which reduce the number of key and value heads, Multi-Head Latent Attention (MLA) compresses the keys and values into a joint latent vector. This compression allows for efficient handling of key-value pairs while maintaining high performance, significantly reducing the KV-cache and improving inference efficiency.

Flash Attention: accelerates the self-attention mechanism by minimizing the memory overhead typical of standard attention calculations. This optimization allows SLMs to process longer sequences more efficiently, enhancing their functionality under strict hardware constraints.

Feedforward Network (FFN): comprises two linear transformations separated by a non-linearity. Generally, ReLU is used as the activation function. In addition to ReLU, activation functions such as GeLU and SiLU are also used in SLMs to improve performance.

3 A Survey on Small Language Models in the era of Large Language Models: Architecture, Capabilities, and Trustworthiness [3]

LLMs with transformer architecture face challeneges with deployment, inference latency and costly fine tuning. SLMs can replace LLMs through innovative architecture and technologies, or assist them as efficient proxy or reward models. Architectures like Mamba and xLSTM address the quadratic scaling of inference with window length in Transformers by enabling linear scaling

Disadvantages of LLMs include:

- common local devices cannot load the large-scale parameters and hard to handle the cache storage;
- huge computational amount cause large inference latency unsuitable for real-time applications;
- they make domain-specific fine-tuning computationally demanding.

The Transformer's reliance on historical key-value pairs causes inefficient inference, facing quadratic computational complexity $O(L^2)$ and high KV cache memory demands for long sequences. Transformer architecture, input tokens receive token embeddings and are processed through self-attention that dynamically weights input importance; this output feeds into a feedforward neural network with an activation function for non-linear processing, and each output is normalized by layer normalization to stabilize learning and facilitate smooth gradient flow.

SLMs enhance LLM security by filtering out potential attack samples, thus mitigating safety risks in conversational AI. Importantly, it is essential to address trustworthiness issues like susceptibility to adversarial attacks and privacy breaches

4 Overview of Small Language Models in Practice [4]

One of the biggest advantages of SLMs is absolute data privacy as all inputs, computations, and outputs are produced and stored locally on one machine without any access to the external world. The main purpose of language model is assign probabilities to sentences in natural language.

Model	Size	Date	Attention	Activation	Layer Norm
$\mathcal{T}$: Phi-4-mini [2]	3.8B	2025.03	GQA	GEGLU	RMSNorm
$\mathcal{T}$: SmolLM-2 [9]	135M, 360M, 1.7B	2025.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: DeepSeek-R1-Distill-Qwen-1.5B [32]	1.5B	2025.01	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: PhoneLM [116]	0.5B, 1.5B	2024.11	MHA	ReLU	RMSNorm
$\mathcal{M}$: Hymba [22]	1.5B	2024.11	SSM+TF	GEGLU	LayerNorm
$\mathcal{T}$: MiniCPM-3 [41]	4B	2024.09	MLA	UNK	RMSNorm
$\mathcal{T}$: Llama-3.2 [5]	1B, 3B	2024.09	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Qwen-2.5 [112]	0.5B, 1.5B, 3B	2024.09	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Phi-3.5-mini [1]	2.7B	2024.09	GQA	GEGLU	RMSNorm
$\mathcal{T}$: DCLM [54]	1.4B	2024.08	MHA	SwiGLU	LayerNorm
$\mathcal{T}$: H2O-Danube3 [77]	0.5B, 4B	2024.07	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Fox-1 [89]	1.6B	2024.07	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: SmolLM [10]	135M	2024.07	GQA	SwiGLU	RMSNorm
	360M		GQA	SwiGLU	RMSNorm
	1.7B		MHA	SwiGLU	RMSNorm
$\mathcal{T}$: Minitron [73]	4B	2024.07	MHA	GELU	LayerNorm
$\mathcal{T}$: Gemma-2 [88]	2B	2024.07	GQA	GELU	RMSNorm
$\mathcal{T}$: Qwen-2 [111]	1.8B, 4B	2024.06	GQA	SwiGLU	RMSNorm
$\mathcal{M}$: Zamba [28]	1.2B	2024.05	SSM+TF	UNK	LayerNorm
$\mathcal{X}$: xLSTM [13]	125M, 350M, 760M, 1.3B	2024.05	LSTM	GELU	LayerNorm
$\mathcal{M}$: Mamba 2 [20]	2.7B	2024.05	SSM	SwiGLU	LayerNorm
$\mathcal{T}$: OpenELM [69]	270M, 450M 1.1B, 3B	2024.04	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: OLMo [29]	1.2B	2024.04	MHA	SwiGLU	RMSNorm
$\mathcal{T}$: Phi-3-mini [1]	3.8B	2024.04	GQA	GEGLU	RMSNorm
$\mathcal{T}$: MiniCPM [41]	1B, 2B	2024.04	GQA	UNK	RMSNorm
$\mathcal{T}$: MobiLlama [90]	0.5B, 1B	2024.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Gemma-1 [87]	2B	2024.02	MQA	GELU	RMSNorm
$\mathcal{T}$: Qwen-1.5 [12]	0.5B	2024.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: StableLM-2-zephyr [14]	1.6B	2024.01	MHA	SwiGLU	LayerNorm
$\mathcal{M}$: Mamba 1 [30]	130M, 370M, 790M, 1.4B, 2.8B	2023.12	SSM	SwiGLU	LayerNorm

Figure 3: Overview of New SLMs

One of the simplest LMs are n-gram models. In n-gram models, text is split into words (called tokens, the process of splitting is called tokenization), and N consecutive tokens form pairs, triples, or other n-tuples of words called n-grams. the frequency of these n-grams is calculated in a training corpus. Using those frequencies, it is possible to assess the "quality" of a sentence and even predict which words come next. It is enough to know only n last words of the sentence and its frequency across whole language in order to predict which word comes next. N-grams can capture the syntax of the text, however knowing just frequencies of sequences of words is not enough for generating "human-like" text. the semantics of the generated text is meaningless

4.1 Transformer

Modern LLMs are made with a Decoder-Only Transformer architecture. There are 4 crucial components in LLMs architecture

4.1.1 Components of Transformer

1. **Embedding Layer:**
   - Converts each input **token** into a dense, fixed-size **vector** (embedding).
   - **Property:** Semantically similar words are represented by vectors that are geometrically **close** to one another in the vector space.
   - **Function:** Provides a low-dimensional, compact, and meaningful representation for the high-dimensional vocabulary.
2. **Transformer Blocks:**
   - The fundamental computational unit, typically **stacked** sequentially within the LLM architecture.
   - **Composition:** Each block primarily consists of a **Multi-Head Attention** layer and a **Feed-Forward Network** (FFN).

- **Function:** Processes the input sequence to transform the embeddings, building complex contextual relationships.

3. **Attention Layer (Multi-Head):**
   - The mechanism that allows all token embeddings to **"attend"** to and interact with one another in the sequence.
   - **Mechanism:** Uses linear transformations (matrix multiplication) involving **Query (Q)**, **Key (K)**, and **Value (V)** matrices.
   - **Function:** It dynamically weights the importance of every other token when processing the current token, thereby forming deep semantic representations.
4. **Embeddings to Logits Conversion:**
   - The final step of converting the output embedding (representing the predicted next word) into a **logits vector**.
   - **Logits:** Raw, unnormalized prediction scores, where the dimension matches the size of the total vocabulary.
   - **Prediction:** A **sampling algorithm** (applied to probabilities derived from the logits, usually via Softmax) selects the most probable next token in the sequence.

In n-gram models tokens resemble words closely, while in LLMs tokenization is a mean of compressing input in order to fit in the context window. The most popular algorithm for tokenization is byte-pair encoding (BPE). The result of BPE is not really interpretable as it may encode input as a set of words, or pairs of words, etc

SLMs are also actively used in fine-tuning, which can bring better performance than using a big production-grade LLM for a specific field of knowledge

4.2 3 main methods of constructing SLMs out of LLMs

- **Pruning:**
  - **Definition:** Involves the systematic **reduction of neurons** or parameters within a model.
  - **Types:**
    1. **Unstructured:** Neurons/weights are removed **uniformly** across the network.
    2. **Structured:** Entire layers or specific components are removed.
  - **Benefit:** Simple method that **significantly reduces model size**.
  - **Risk:** Over-pruning can lead to severe degradation in model performance [**?**].
- **Knowledge Distillation:**
  - **Mechanism:** Involves two models: a large **Teacher** model (LLM) and a smaller **Student** model (SLM).
  - **Training:** The Student model is trained not just on the original data labels, but primarily on the **soft outputs** (probabilities/logits) of the Teacher model.
  - **Benefit:** Effectively **transfers the performance** and learned knowledge of a bigger, more complex model to a smaller, more efficient one.
  - **Limitation:** Still requires the full LLM (Teacher) to be run initially to generate the training outputs [**?**].
- **Quantization:**
  - **Definition:** Achieved by converting the high-precision floating-point numbers used for weights and activations into **simplified low-precision number formats**.
  - **Popular Levels:** Includes 16-bit floats (half-precision), 8-bit integers, 4-bit, and even 2 and 1 bits [**?**].
  - **Benefit: Significantly reduces model size** (up to $4\times$ for 8-bit) and accelerates inference times.
  - **Requirement:** May necessitate **specialized hardware** capable of efficiently processing these low-precision formats.

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